ORLANDO HEALTH®
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# DEPARTMENT GUIDELINE

| Guideline Title: **Pediatric Cardiogenic Shock** | Guideline Number: **GUID-3203-PED006** |
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| Issue date: 09/09/2014 | Replaces Dept. Policy: |
| Revision dates: 08/11/2026 | Developed by: Air Care Team |
| Approved by: Dr. Danielle DiCesare, MD<br>Air Care Team Medical Director | Approved by: |
| Signature: | Signature: |
| Department Numbers: 3203 | |

I. **PURPOSE:** Restoration of inadequate cardiac output to improve perfusion by expanding intravascular volume, improving contractility, and treating the underlying cause.

II. **PATHOPHYSIOLOGY:** Common causes of cardiogenic shock include congenital heart disease, myocarditis, cardiomyopathy, arrhythmias, sepsis, drug toxicity, or myocardial injury (trauma). Marked tachycardia, high SVR, and decreased cardiac output characterize cardiogenic shock. Pulmonary venous congestion leads to pulmonary edema and increased work of breathing. Typically, intravascular volume is normal or increased unless a concurrent illness causes hypovolemia.

III. **DEFINITIONS:**

IV. **DEPARTMENT GUIDELINE:**

A. Assess and manage airway, breathing, and circulation. Monitor ETCO2 if intubated. Optimize oxygen content.

B. Assess for airway patency, anxiety/agitation, heart rate, pulse quality, skin perfusion, respiratory distress (retractions, grunting, and use of accessory muscles), signs of fluid overload, hypotension, and level of consciousness.

C. Obtain the largest IV access for the patient's size. 2 points of IV access are preferred.

D. If unable to obtain IV, obtain IO access per *GUID3203-PR013 – Intraosseous Infusions – EZIO.*

E. Manage cardiac arrythmias (supraventricular/ventricular tachycardia/bradycardia) prior to fluid resuscitation. Consider betablocker or calcium channel blocker poisoning.

F. If signs of pulmonary vascular congestion are present, withhold fluid bolus and consult with physician for order to administer **Furosemide 1-2mg/kg IV** max single dose 20 mg**.**

G. Rapidly providing volume resuscitation of cardiogenic shock in the setting of poor myocardial function can aggravate pulmonary edema and further impair myocardial function. Gradually provide volume resuscitation for cardiogenic shock. If no evidence of pulmonary vascular congestion or fluid overload:

   1. Resuscitate with 0.9% NS or Lactated Ringers 5-10 ml/kg warm NS or LR bolus over 10-20 minutes.
      a. Repeat bolus if patient remains hypo-perfused unless signs of pulmonary vascular congestion or fluid overload develop.
      b. Warm fluids per *GUID3203-PR026 – Warming of Blood Product & Intravenous Fluids*
   2. For patients failing fluid boluses, consider initiating inotropic support early.

   a. **Dobutamine infusion at 2 - 20 mcg/kg/minute**. –OR-
   b. **Epinephrine infusion at 0.02 - 1 mcg/kg/min.**
   c. **Milrinone infusion at 0.25 - 0.75mcg/kg/min.**

H. Most children with cardiogenic shock benefit from a vasodilator (if blood pressure is adequate) to decrease systemic vascular resistance (SVR) and increase cardiac output and tissue perfusion. Obtain consultation from sending or receiving facility.

I. If the blood glucose level is less than 60, treat per *GUID3203-M021 – Hypoglycemia*.

J. Monitor urine output by indwelling urinary catheter if available. Titrate resuscitation to 1ml/kg/hr.

K. Always consider the latest recommendations from the American Heart Association.

II. **DOCUMENTATION:**

A. EMS Charts

III. **REFERENCES:**

A. Pediatric Advanced Life Support Manual, 2020. American Heart Association.